

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
ACADEMIC ASSIGNMENT

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CO5 - AT 3
MOCK INTERVIEW

Network Performance and Simulation Tools

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Declaration

I hereby declare that this assignment is my own original work, prepared under the guidance of my course faculty, and has been submitted in accordance with the academic requirements of the course.

1. Analyze how network congestion affects packet delay and packet loss.

Answer: Network congestion occurs when offered traffic exceeds the capacity of a link or router. Packets then wait longer in queues, increasing queuing delay. If router buffers become full, new packets are dropped, causing packet loss and possible retransmissions. Thus, severe congestion increases latency, reduces throughput, and degrades user experience.

2. Compare the impact of star and mesh topologies on network reliability and fault tolerance.

Answer: In a star topology, failure of one end-device link affects only that device, but failure of the central switch can disrupt the entire network. A mesh topology provides multiple paths between nodes, so traffic can be rerouted when a link fails. Therefore, mesh offers higher reliability and fault tolerance, although it requires more links, ports, and cost.

3. Why does increasing the number of routers between source and destination increase end-to-end delay? Explain logically.

Answer: Every router introduces processing delay while examining headers, queuing delay when traffic is busy, and transmission delay when sending the packet onto the next link. With more hops, these delays accumulate. Therefore, a path containing many routers usually has greater end-to-end delay than a shorter path under similar conditions.

4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic.

Answer: QoS identifies and prioritizes delay-sensitive traffic such as voice and video over less time-critical traffic. Techniques such as traffic classification, priority queuing, bandwidth reservation, and shaping reduce delay, jitter, and packet loss for real-time applications. This helps maintain clear VoIP calls and smooth video even when the network is busy.

5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication.

Answer: Fiber optic cable supports much higher bandwidth, longer transmission distance, lower signal attenuation, and immunity to electromagnetic interference. It is therefore suitable for high-speed backbones and data centers. Its disadvantages are higher initial equipment and installation cost and more specialized handling compared with copper.

6. Analyze how network segmentation using VLANs improves performance and security.

Answer: VLANs divide a physical switched network into smaller logical broadcast domains. This reduces unnecessary broadcast traffic and can improve performance. Security is improved because departments or user groups are isolated, and communication between VLANs can be controlled using Layer 3 routing, ACLs, or firewalls.

7. Compare the effects of TCP and UDP on network performance for different applications.

Answer: TCP provides reliable, ordered delivery using acknowledgements, retransmissions, flow control, and congestion control, but these features add overhead and delay. UDP has lower overhead and sends data without guaranteeing delivery or order. TCP suits web, email, and file transfer, while UDP is preferred for real-time voice, video, gaming, and DNS where low latency is important.

8. Why is load balancing important in large enterprise networks? Analyze its impact on performance and availability.

Answer: Load balancing distributes traffic across multiple servers or network paths instead of overloading a single resource. It improves throughput, response time, and resource utilization. It also increases availability because traffic can be redirected to healthy resources when one server or path fails.

9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization.

Answer: Routing updates are necessary to maintain correct path information, but excessive updates consume bandwidth and router CPU resources. Very frequent changes may also cause route instability and temporary packet loss. Efficient routing protocols reduce unnecessary updates through triggered changes, timers, route summarization, and stable convergence mechanisms.

10. Evaluate the trade-off between network security measures such as firewalls and encryption and overall network performance.

Answer: Firewalls inspect packets and encryption performs additional processing, so both may introduce latency and consume CPU or hardware resources. However, they are essential for confidentiality, access control, and threat protection. The best design balances security and performance by using efficient rules, hardware acceleration, proper placement, and suitable encryption methods.

11. A network has high delay—how will you identify and fix the issue?

Answer: I would first measure latency using ping and use traceroute to identify the hop where delay increases. Then I would check interface utilization, errors, queue lengths, routing paths, and device CPU usage. Depending on the cause, I would remove congestion, correct faulty links or configurations, apply QoS, optimize routing, or increase bandwidth.

12. If throughput is low despite high bandwidth, what could be the reasons?

Answer: Low throughput can result from packet loss, high RTT, congestion, protocol overhead, small TCP windows, server limitations, interface errors, duplex mismatch, or overloaded network devices. Available bandwidth is only the maximum capacity; actual throughput depends on the complete end-to-end path and the efficiency of hosts and protocols.

13. How would you reduce packet loss in a congested network?

Answer: I would identify overloaded links, remove unnecessary traffic, apply QoS and traffic shaping, and distribute traffic using load balancing. Increasing bandwidth or upgrading bottleneck devices may also be required. Proper queue management and congestion-control mechanisms help prevent buffers from overflowing and dropping packets.

14. A VoIP call is breaking frequently—how would you troubleshoot?

Answer: I would measure packet loss, latency, and jitter between the VoIP endpoints and check available bandwidth. I would also inspect Wi-Fi quality, interface errors, routing, and QoS configuration. Prioritizing RTP/voice traffic, reducing congestion, using a suitable codec, and correcting unstable links can improve call quality.

15. Given a network with high jitter, suggest improvements.

Answer: High jitter can be reduced by prioritizing real-time traffic with QoS, reducing congestion, and using stable low-latency routes. Traffic shaping and appropriate buffering can smooth irregular packet arrival. Replacing unreliable links and avoiding overloaded wireless channels can also improve consistency.

16. How will you simulate a congested network in Packet Tracer?

Answer: I would create a topology with multiple hosts sending traffic through a common low-bandwidth link or router. I would generate simultaneous application traffic or PDUs and use Simulation mode to observe packet movement, queueing, and drops. By increasing traffic or reducing link capacity, the effect of congestion can be demonstrated and compared.

17. If packets are delayed in a router, what steps will you take?

Answer: I would check interface utilization, queue status, packet drops, CPU usage, routing-table entries, and interface errors. If the router is congested, I would prioritize critical traffic using QoS, balance traffic across alternate paths, reduce unnecessary traffic, or increase link capacity. A faulty or overloaded router may require configuration correction or upgrade.

18. How would you optimize a network for online gaming?

Answer: Online gaming needs low latency, low jitter, and minimal packet loss. I would prefer a wired Ethernet connection, reduce background downloads, apply QoS to prioritize gaming traffic, avoid congested links, and use an efficient route to the game server. A stable connection is usually more important than simply having very high bandwidth.

19. Design a small network and explain how you will measure its performance.

Answer: I would connect several PCs to a switch, connect the switch to a router, and include a server or another LAN as the destination. I would generate traffic and measure delay, RTT, throughput, packet loss, and jitter using ping, traffic tests, and Packet Tracer Simulation mode. The results would be recorded under normal and high-load conditions for comparison.

20. If RTT is very high, how will you diagnose the issue?

Answer: I would first ping the destination and then use traceroute to locate the hop causing the largest increase in delay. I would check congestion, routing changes, long physical distance, wireless interference, interface errors, and ISP-related delay. After identifying the cause, I would optimize routing, reduce congestion, repair the link, or escalate an external provider issue.

21. Explain network performance to a non-technical person.

Answer: Network performance describes how quickly and reliably information travels between devices. It is similar to road traffic: bandwidth is the road capacity, delay is travel time, congestion is a traffic jam, and packet loss is like some deliveries never arriving. A good network transfers data quickly, consistently, and with very few losses.

22. Describe throughput using a real-life example.

Answer: Throughput is the amount of useful data that is actually delivered in a given time. It is like a water pipe: the pipe may have a large maximum capacity, but throughput is the amount of water that actually comes out each second. Network congestion or loss can make actual throughput lower than the available bandwidth.

23. Present how packet loss affects video streaming.

Answer: Packet loss means some parts of the video data fail to reach the receiver. This can cause buffering, freezing, pixelation, missing audio, or a reduction in video quality. If the application retransmits lost data, playback may be delayed; if it does not, the missing information may appear as visible or audible distortion.

24. Explain delay types clearly in under 1 minute.

Answer: The four common delay types are processing, queuing, transmission, and propagation delay. Processing delay is the time a device takes to examine a packet; queuing delay is waiting time in a buffer; transmission delay is the time required to place all packet bits onto a link; and propagation delay is the time the signal takes to travel through the medium.

25. Describe Cisco Packet Tracer in a professional way.

Answer: Cisco Packet Tracer is a network simulation and visualization tool used to design, configure, test, and troubleshoot virtual networks. It supports routers, switches, end devices, protocols, and simulation of packet flow. It is useful for learning networking concepts and testing configurations without requiring a full physical laboratory.

26. Explain QoS in simple and technical terms.

Answer: In simple terms, QoS gives important traffic a fast lane when the network is busy. Technically, Quality of Service classifies traffic and applies mechanisms such as marking, priority queuing, shaping, policing, and bandwidth allocation. Its goal is to control delay, jitter, packet loss, and bandwidth for applications with different requirements.

27. How would you present your simulation results to a client?

Answer: I would begin with the network objective and topology, then explain the test conditions and metrics measured. I would present clear tables or charts for throughput, delay, jitter, and packet loss, comparing normal and congested conditions. Finally, I would explain the business impact and provide practical recommendations rather than only showing raw technical data.

28. Explain jitter using a real-world analogy.

Answer: Jitter is variation in the arrival time of packets. It is like a bus service where buses are scheduled every five minutes but actually arrive after two minutes, then eight minutes, then three minutes. Even if the average travel time is acceptable, irregular arrival causes problems for continuous services such as voice and video.

29. Describe how you tested network performance in a lab.

Answer: In a lab, I would build a topology, configure IP addressing and routing, verify connectivity, and then generate traffic between endpoints. I would use ping and simulation tools to observe RTT, packet delivery, and protocol behavior, and record throughput, packet loss, delay, and jitter where available. I would repeat tests under different loads and compare the results.

30. Explain the importance of performance metrics in one structured answer.

Answer: Performance metrics provide measurable evidence of how well a network is operating. Bandwidth and throughput show capacity and actual data delivery, delay and RTT show responsiveness, jitter shows consistency, and packet loss shows reliability. Together, these metrics help detect bottlenecks, compare designs, plan capacity, verify QoS, and improve user experience.

31. Introduce yourself and explain your knowledge of network performance.

Answer: I am a student with a strong interest in computer networks and practical network analysis. I understand important performance metrics such as bandwidth, throughput, delay, jitter, RTT, and packet loss, and I can relate these metrics to congestion, QoS, routing, and topology design. I am also comfortable using simulation tools to observe and troubleshoot network behavior.

32. Why are you interested in networking and simulation tools?

Answer: I am interested in networking because reliable communication is essential for almost every modern digital service. Simulation tools allow me to design networks, test configurations, study packet flow, and troubleshoot problems safely without needing expensive physical equipment. They help connect theoretical concepts with practical behavior.

33. What makes you confident in analyzing network performance?

Answer: My confidence comes from following a structured approach: first measure the relevant metrics, then identify where the problem occurs, analyze the likely cause, apply a solution, and test again. Understanding throughput, delay, jitter, packet loss, routing, and congestion helps me explain performance problems logically rather than guessing.

34. Explain a concept you are very strong in related to networking.

Answer: One concept I am strong in is Quality of Service. QoS classifies network traffic and gives suitable priority or bandwidth to applications based on their requirements. For example, voice traffic can be given higher priority than a large file download so that delay and jitter remain low during congestion.

35. How would you handle a question you don't know?

Answer: I would not guess confidently. I would clarify the question, explain the related concepts that I do know, and logically work toward a possible answer. If I still cannot confirm it, I would state that clearly and verify the correct information from reliable documentation afterward.

36. Describe a situation where you solved a technical problem.

Answer: A suitable example is a lab network showing high delay during heavy traffic. I would measure the affected path, identify the congested link, verify queueing and utilization, and then apply QoS or reduce competing traffic. I would repeat the test afterward to confirm that delay and packet loss had improved.

37. Why should we select you for a networking role?

Answer: I have a good foundation in networking concepts and I approach problems systematically using measurable performance data. I can work with topology design, routing, VLANs, QoS, troubleshooting, and simulation tools, and I am willing to learn new technologies continuously. I focus on finding practical solutions rather than only memorizing theory.

38. What challenges did you face while learning Packet Tracer?

Answer: Common challenges include choosing correct cables and interfaces, configuring IP addresses and routing correctly, understanding why packets fail, and interpreting Simulation mode events. I improve by building small topologies first, verifying each layer step by step, and using repeated testing to locate configuration mistakes.

39. How do you improve your technical skills regularly?

Answer: I improve by studying core concepts, practicing configurations and troubleshooting in simulation tools, solving networking problems, and reviewing mistakes after each lab. I also compare test results and learn how changing bandwidth, routing, QoS, or topology affects performance. Regular hands-on practice makes the concepts easier to remember and apply.

40. Demonstrate confidence while explaining a complex concept simply.

Answer: For example, packet loss can be explained like sending 100 letters through a delivery system and only 95 reaching the destination. The missing five letters represent a 5% packet loss. In a network, too much loss can cause retransmissions, buffering, broken calls, or poor video quality, so reducing congestion and fixing unreliable links is important.